



Building a Spanish conversational system with router-controlled RAG

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Motivation & research questions

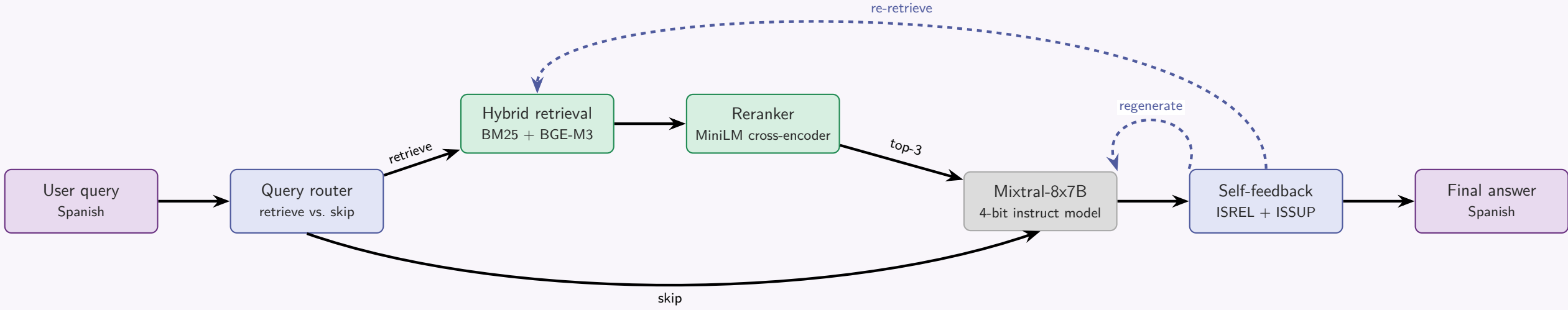
Retrieval-Augmented Generation (RAG) grounds factual answers in a corpus, but **always retrieving is wasteful**: conversational turns (greetings, jokes, emotional support) neither need nor benefit from corpus lookup.

This project builds a **Spanish conversational assistant** that first classifies each query (**retrieve** or **skip**) then generates an answer with Mixtral (Jiang et al., 2024), with an optional **self-feedback** loop inspired by Self-RAG (Asai et al., 2023).

Research questions:

- RQ1: Can a learned router separate factual questions that need RAG from open conversational turns?
- RQ2: Does routing preserve RAG answer quality while reducing latency?
- RQ3: Which router training strategy works best with fewest examples?
- RQ4: Does self-feedback repair unsupported or incorrect RAG answers?

System pipeline



The router is the control point: it decides whether the expensive retrieval branch is worth running.

Datasets & data construction

Initial dataset with dataset labels

Source	Split	N samples	Label
SQAC	Router train	15,036	retrieve
ChatSubs	Router train	15,036	skip
SQAC	Router val	1,779	retrieve
ChatSubs	Router val	1,664	skip
SQAC	Test	85	retrieve
MS Chitchat	Test	85	skip

Generated oracle-labelled train sets

Source	Subset	Total	Per class
SQAC	Oracle-500	1,000	500
SQAC	Oracle-64	128	64

Oracle labels are produced by generating RAG-based output (*retrieve* class) and a RAG-skipped output (*skip* class). The output that obtains a higher BERT-F1 score against the gold answer is the oracle label of that sample.

Retrieval setup

- 5,601 SQAC passages
- BM25 + normalised BGE-M3 dense vectors.
- MiniLM reranker outputs top-3 passages
- **Critic prompt asks Mixtral for feedback**: `isrel` (context relevance) and `issup` (answer support).

Router training strategies

Router approaches

- **Frozen LR**: frozen multilingual MPNet embeddings + logistic regression. Trains in seconds. Strong baseline.
- **SetFit**: contrastive few-shot fine-tuning. Learns from as few as 8 examples per class.
- **Fine-tune**: full encoder fine-tuning with HuggingFace Trainer (5 epochs, early stopping).

Metrics

- Routing: pre-feedback router accuracy and post-feedback effective accuracy.
- Answer quality: token F1 and multilingual BERTScore F1.
- Cost: mean latency and pre/post-feedback retrieval rate.
- Oracle policy: retrieve iff RAG-based output beats RAG-skipped in BERTScore F1.

Finding 1: Routing preserves quality at lower cost

Model	Retrieval rate ↓	Latency ↓	Token-F1 ↑	BERT-F1 ↑
Always RAG	100.0%	13.94s	7.88	61.10
Never RAG	0.0%	6.64s	4.57	58.21
Frozen LR router	48.2%	10.33s	7.39	60.81
SetFit router	48.8%	10.05s	8.12	61.22
Fine-tuned router	50.6%	9.98s	7.64	61.09
Oracle SetFit-500	47.6%	9.82s	6.98	59.78
Oracle SetFit-64	44.1%	10.10s	6.03	59.68

Answer quality. SetFit is the only router model that beats Always RAG. It performs slightly better in both token F1 and BERTScore while retrieving on only about half of the test set.

Latency. All router models outperform the Always RAG baseline. SetFit reduces mean latency from 13.94s to 10.05s, which is a **27.9% reduction** relative to Always RAG, while the fine-tuned router is slightly faster at 9.98s.

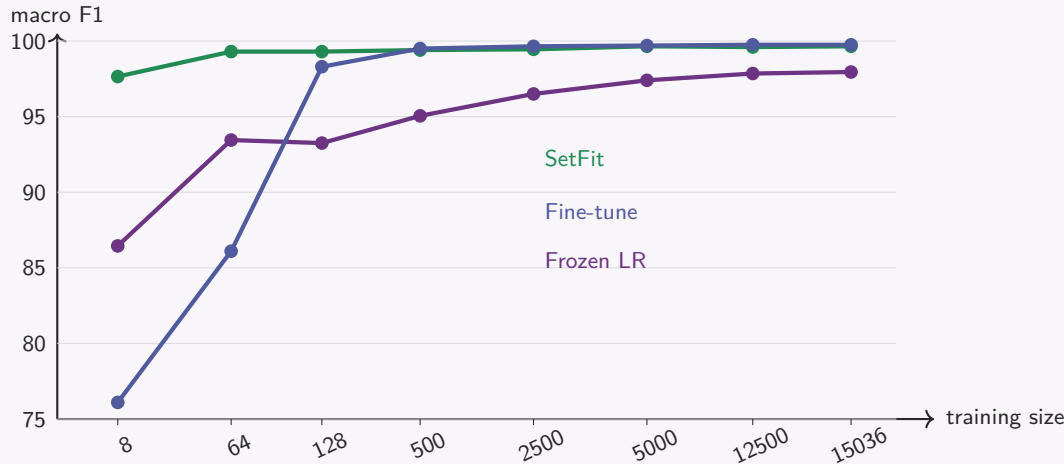
Finding 3: Dataset accuracy does not correspond to oracle accuracy

Model	Dataset accuracy ↑	Oracle accuracy ↑
Always RAG	50.00	69.41
Never RAG	50.00	30.59
Frozen LR router	97.06	67.06
SetFit router	98.82	69.41
Fine-tuned router	99.41	69.41
Oracle SetFit-500	52.94	57.65
Oracle SetFit-64	52.94	49.41

Subset	Oracle retrieval
SQAC	69.4%
Chitchat	0.0%
Overall	34.7%

Router accuracy evaluates the first binary retrieve/skip decision, before any self-feedback can discard irrelevant context.
Main insight: The learned routers solve the dataset-label task, but high router accuracy alone does not guarantee the best answer quality or oracle policy.

Finding 2: The router learns the dataset labels quickly



Learning curve. SetFit reaches **97.6% macro F1 with only 8 examples per class**. Full fine-tuning catches up after 500 examples per class and reaches the highest validation score at 12,500 examples per class.

Finding 4: Self-feedback (SF) improves oracle accuracy

Model	Retrieval rate ↓			Latency ↓			Dataset acc. ↑			Oracle acc. ↑		
	NoSF	SF	Δ	NoSF	SF	Δ	NoSF	SF	Δ	NoSF	SF	Δ
Always RAG	100.0	61.8	-38.2	6.21s	13.94s	+7.72s	50.00	61.18	+11.18	69.41	77.65	+8.24
Frozen LR router	48.2	34.1	-14.1	5.82s	10.33s	+4.51s	97.06	84.12	-12.94	67.06	72.94	+5.88
SetFit router	48.8	35.9	-12.9	5.54s	10.05s	+4.51s	98.82	85.88	-12.94	69.41	76.47	+7.06
Fine-tuned router	50.6	37.6	-12.9	5.53s	9.98s	+4.45s	99.41	87.65	-11.76	69.41	75.29	+5.88
Oracle SetFit-500	47.6	32.4	-15.3	6.77s	9.82s	+3.05s	52.94	58.82	+5.88	57.65	57.65	+0.00
Oracle SetFit-64	44.1	22.9	-21.2	6.75s	10.10s	+3.35s	52.94	57.65	+4.71	49.41	49.41	+0.00

Main effect: irrelevant retrieval is filtered out. In Always RAG, self-feedback lowers effective retrieval from 100% to 61.8%.

Self-feedback changes the policy target. For dataset-label routers, dataset accuracy drops after self-feedback, but oracle-policy match rises by 5–7 points because some retrieved contexts are not useful.

Qualitative analysis

Good skip: conversational query

Query: “Dime si puedes estornudar”
Route: skip retrieval
Answer: “No, no puedo estornudar porque soy un asistente digital...”
Effect: direct answer without forcing irrelevant retrieved context.

Good retrieve: factual QA

Query: “¿Qué cargos ocupa Jesús Gil?”
Route: retrieve SQAC passages
Answer: “alcalde de Marbella y presidente del Atlético de Madrid”
Effect: retrieved context supports a concise gold-like answer.

Good routing but unsuccessful retrieval

Query: “¿Qué es el SITE?”
Gold: “grupo de inteligencia”
Route: retrieve → fallback
Answer: “SITE se puede referir a Society of Inventors and Technical Executives...”
Effect: retrieval failed to recover relevant SQAC evidence.

Main insight: the router reliably separates conversational and factual queries. The main bottleneck is retrieval quality, not the routing decision itself.

Take-aways

1. **A learned router is worthwhile:** same answer quality while achieving a **considerable reduction in computational cost** (retrieval rate 100% to 48.8%) **and temporal cost** (27.9% lower latency) relative to Always RAG.
2. **SetFit is the practical router choice.** It trains fast and effectively (**99% validation accuracy with only 64 examples per class**). Fine-tuning achieves same performance at larger training sizes. Frozen LR lags behind.
3. **Oracle-label training is harder than dataset-label training.** Routers trained on oracle labels underperform the dataset-label routers on oracle accuracy.
4. **Self-feedback boosts oracle accuracy**, especially in the Always-RAG model. Still, **SetFit** is the **best model** since it achieves **almost the same oracle accuracy as Always-RAG with lower retrieval rate and latency**.

Limitations & future work

Limitations

1. Oracle labels are model dependent (no solution, each LLM is pretrained on different data).
2. Fallback can produce ungrounded answers (solution in future work).

Future work

1. Extend the oracle labels corpus size.
2. Add **grounding-aware abstention**: if retrieval confidence or critic support is too low, refuse to answer.

References:

Asai et al. (2023). *Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*. arXiv:2310.11511
Jiang et al. (2024). *Mixtral of Experts*. arXiv:2401.04088